

EARLY JET AND TURBOPROP AIRLINERS

THE BASIS OF THE AIRLINE BUSINESS is not technical innovation or elegance of design, but the economic equation of passenger numbers against operating costs. Jet engine technology did not at first seem an attractive proposition to airlines because of high fuel and maintenance costs and low passenger payload. Turbojets were developed in the postwar period to provide some of the comfort and speed advantages of jets, but with more economical operating costs. The Boeing 707 was the breakthrough jet that could carry enough passengers fast enough to make it profitable on long-distance routes. Its success in the 1960s left plenty of room at the bottom for less powerful airliners that could use smaller airports or operate on shorter routes where it was important to perform economically at slower speeds. The BAC 111, one of the few commercially successful British aircraft, catered for this market, as did Douglas's DC-9, but once again it was Boeing that tailored airliners most precisely to the needs of its customers with the 727 and 737.



CARAVELLE

The Caravelle was a trend-setting design – its rear-mounted engines were much imitated – but it did not carry enough passengers to operate profitably. See pages 384–85.

BAC 111

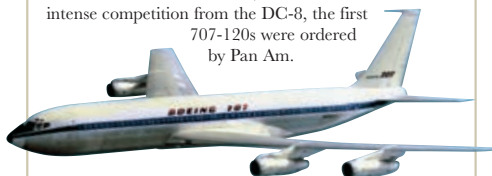


The BAC 111, which entered service in 1965, was based on a similar design to the Sud-Aviation Caravelle, the world's first short-haul jet airliner. An unusual feature of the design, copied by Hunting aircraft (part of BAC) was the tail mounting of the twin engines.

ENGINE 2 x 12,000lbst (5,690kgp) thrust Rolls-Royce Spey turbofan	
WINGSPAN 93ft 6in (28.5m)	LENGTH 107ft (32.6m)
CRUISING SPEED 541mph (871kph)	
PASSENGERS 97	CREW 2

Boeing 707-120

The development of the 707 was a major gamble by Boeing. Determined to break the Lockheed/Douglas monopoly, Boeing bet half the total value of the company on building the first dual-role military tanker/civil airliner. In 1955, in the face of intense competition from the DC-8, the first 707-120s were ordered by Pan Am.



ENGINE 4 x 13,500lb (6,120kg) Pratt & Whitney JT3C-6 turbojet	
WINGSPAN 130ft 10in (39.9m)	LENGTH 144ft 6in (40m)
CRUISING SPEED 571mph (919kph)	
PASSENGERS 121–179	CREW 4

Boeing 727



Although Boeing built a shorter ranged 707, it could not use the smaller airports where passengers also demanded the speed and comfort of jet travel. The answer appeared in 1963 as the rear-mounted trijet 727, with its complex multiflap wing married to a spacious fuselage with the same diameter as the 707. The design proved incredibly successful, becoming the world's best-selling jet transport with 1,831 built when production ended in 1984.

ENGINE 2 x 16,000lbst (7,260kgp) P&W JT8D-17 turbofan	
WINGSPAN 108ft (32.9m)	LENGTH 153ft 2in (46.7m)
CRUISING SPEED 599mph (964kph)	
PASSENGERS 134–189	CREW 3

Boeing 737-200



With its short, fat fuselage and wing-mounted engines, the 737 (737-100 prototype shown) lacked the elegance of its DC-9 and BAC 111 short-range competitors, but the "big jet" feeling was very popular with passengers. The success of the 737 far outstripped its predecessors and by early 2002, orders for the 737 family exceeded 5,000.

ENGINE 2 x 15,500lbst (7,040kgp) P&W JT8D-9A turbofan	
WINGSPAN 93ft (28.4m)	LENGTH 100ft 2in (30.5m)
CRUISING SPEED 575mph (927kph)	
PASSENGERS 115–130	CREW 2

de Havilland D.H.106 Comet 4B

The Comet 1 was the first jet airliner. In 1952, 10 were delivered to British Overseas Airways Corporation. They cut flying times by half, but a series of disastrous accidents led to the grounding of all Comet 1s by 1954. Lengthy investigations showed the accidents were caused by metal fatigue, a little known phenomenon at that time. The improved Comet 4 had longer range. On October 4, 1958, two Comets, one eastbound and another westbound, made the first jet-powered passenger flights across the Atlantic.

ENGINE 4 x 10,500lbst (4,760kgp) thrust R-R Avon 524 turbojet	
WINGSPAN 114ft 10in (35m)	LENGTH 111ft 10in (34m)
CRUISING SPEED 503mph (809kph)	
PASSENGERS 60–81	CREW 4

